L2.2 The Mean Value Theorem

§3.2 — what you need to know

The six theorems

① **IVT** — A continuous curve hits every value between its endpoint heights.

② **EVT** — On a closed interval, a continuous function reaches a highest and a lowest value.

③ **Fermat** — At a local max or min where f' exists, $f'(c) = 0$.

④ **Closed Interval Method** — Evaluate f at the critical numbers and both endpoints; take the largest and smallest.

⑤ **Rolle** — Equal endpoint heights force a horizontal tangent between them.

⑥ **MVT** — Somewhere between the endpoints, the tangent is parallel to the secant.

All six are used the same way: name it, verify each hypothesis one at a time, then use the conclusion. An unchecked hypothesis is a proof that proves nothing.

ROLLE'S THEOREM AND THE MEAN VALUE THEOREM

Rolle. If f satisfies:

— continuous on $[a, b]$
— differentiable on (a, b)
— $f(a) = f(b)$

$\boxed{\exists\, c \text{ on } (a, b) \text{ s.t. } f'(c) = 0}$

MVT. If f satisfies:

— continuous on $[a, b]$
— differentiable on (a, b)

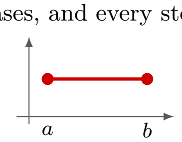
$\boxed{\exists\, c \text{ on } (a, b) \text{ s.t. } f'(c) = m_{\text{sec}}}$

"MVT is a slanted Rolle"

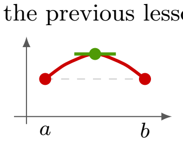
Drop the third hypothesis and the horizontal tangent tilts with the ends. $m_{\text{sec}} = \dfrac{f(b) - f(a)}{b - a}$ is the secant slope you already know, so the conclusion reads *somewhere strictly inside, the tangent is parallel to the secant.* **Rolle is the case** $m_{\text{sec}} = 0$, and the conclusion lives on (a, b), never $(a, b]$.

DERIVATION OF ROLLE'S THEOREM

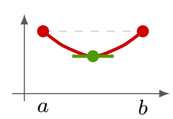
Three cases, and every step is a theorem from the previous lesson.



① f **constant.** $f' = 0$ everywhere on (a, b) — any c works.



② f **goes up. EVT** gives a Max; the equal ends put it *inside*; **Fermat** gives $f'(c) = 0$.



③ f **goes down.** The same argument with a min.

Both hypotheses get spent in case ②: continuity on the closed interval is what EVT needs, differentiability on the open interval is what Fermat needs.

EXACTLY ONE REAL ROOT

Two claims, two theorems. **At least one** is **IVT** — f continuous, and f changes sign across the interval. **At most one** is **Rolle**, by contradiction: two roots would make $f(a) = f(b)$, so Rolle would hand you a c with $f'(c) = 0$, and if f' is never zero that is impossible.

"assume something you don't know is true and see what happens — if it leads to a contradiction, the thing you assumed is wrong"

The template, and it works on any *at most one* claim:

① Assume the **opposite** of the claim — here, that a second root exists.

② Apply a theorem whose hypotheses the assumption just handed you.

③ Push to a statement that contradicts something already known — usually a fact about f'.

④ The assumption was false. The claim stands.

FUNCTIONS WITH THE SAME DERIVATIVE

$$\boxed{f'(x) = g'(x) \text{ on an interval} \implies f(x) = g(x) + c}$$

Corollary 7, and it is one line from the Mean Value Theorem: if $h' = 0$ on an interval then h is constant there, so apply that to $h = f - g$.

Later this chapter the question runs backwards — given f, find something whose derivative is f — and the answer gets written $F(x) + C$. **This corollary is why that C sweeps up *every* such function.** Without it, $+C$ is a habit; with it, the family is provably complete.

The word *interval* is a hypothesis, not decoration — on a domain made of two separate intervals the conclusion fails.

WATCH OUT

✗ ~~Rolle needs $f(a) = f(b) = 0$~~ → **it needs them equal; zero is irrelevant**

✗ ~~the theorem hands me the number c~~ → **$\exists$ is existence; solving $f'(x) = m_{\text{sec}}$ gives the number**

✗ ~~c is unique~~ → **at least one, never exactly one — two is common**

✗ ~~$c = b$ solves the equation, so it counts~~ → **the conclusion is on (a, b) — an endpoint is rejected**

✗ ~~f must be differentiable on $[a, b]$~~ → **only on the open (a, b) — $4x - \sqrt{x}$ on $[0, 4]$ qualifies with no $f'(0)$**

✗ ~~there is no c, so the theorem is false~~ → **a failed hypothesis is the theorem working**